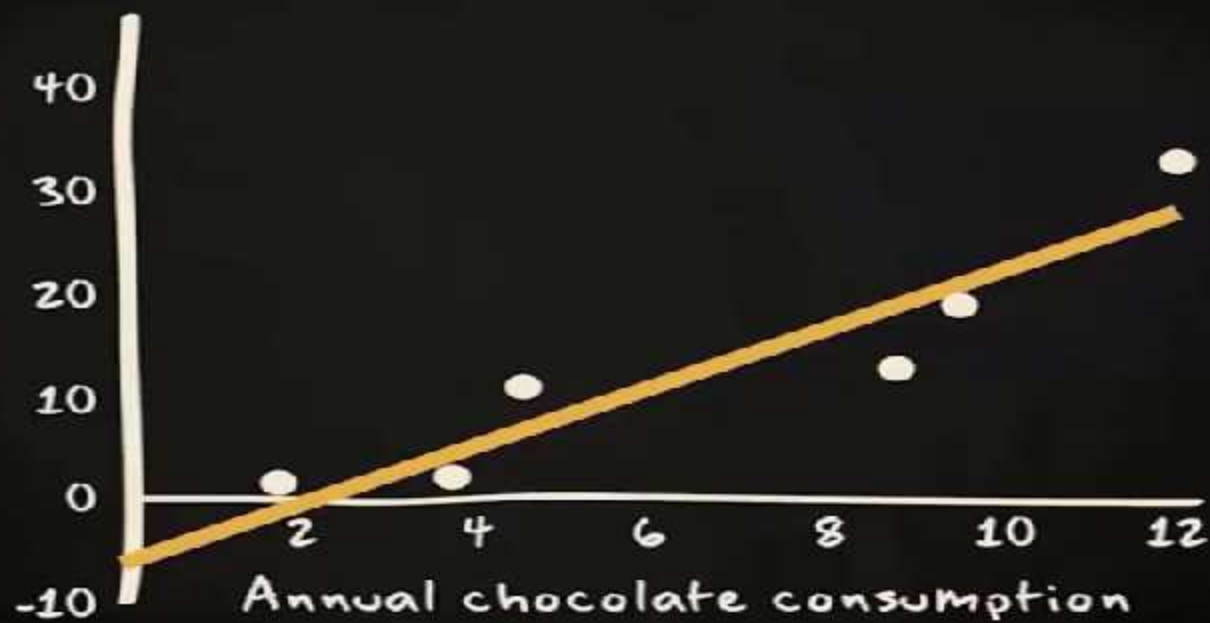


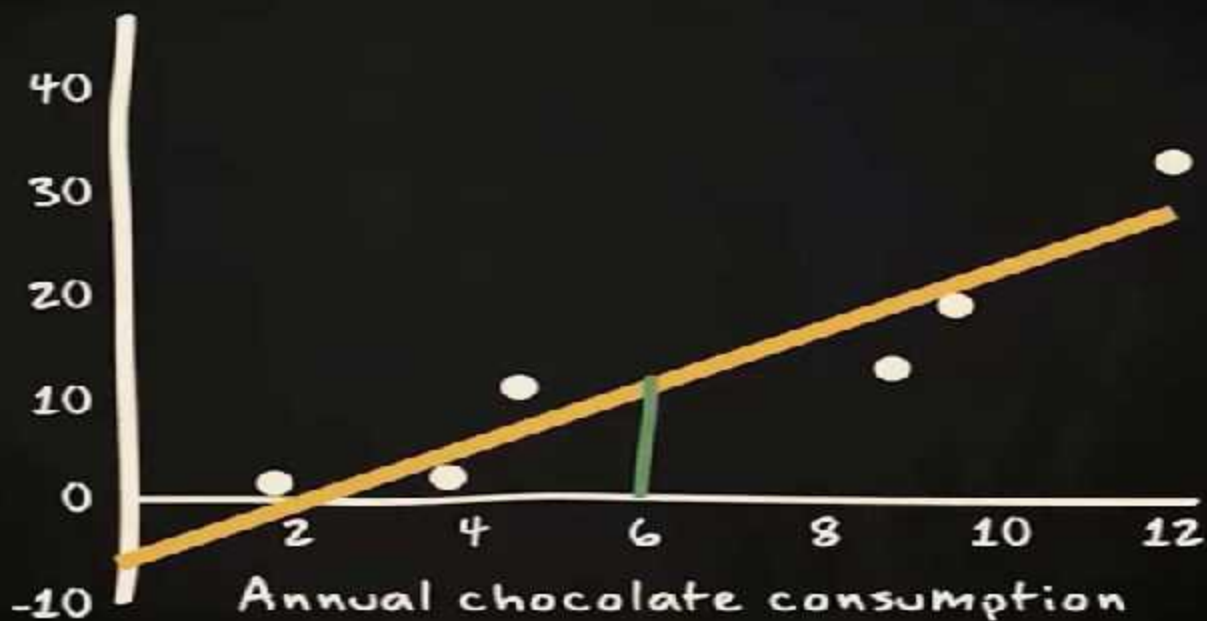


Number of Nobel prize winners
per 10 million people



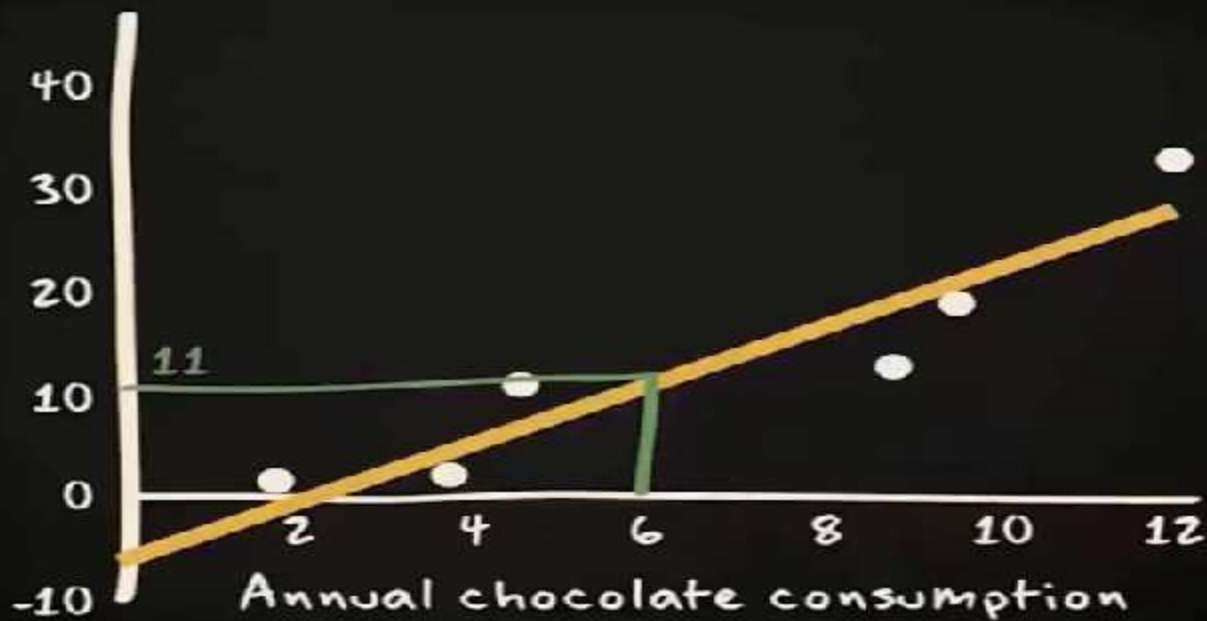


Number of Nobel prize winners
per 10 million people





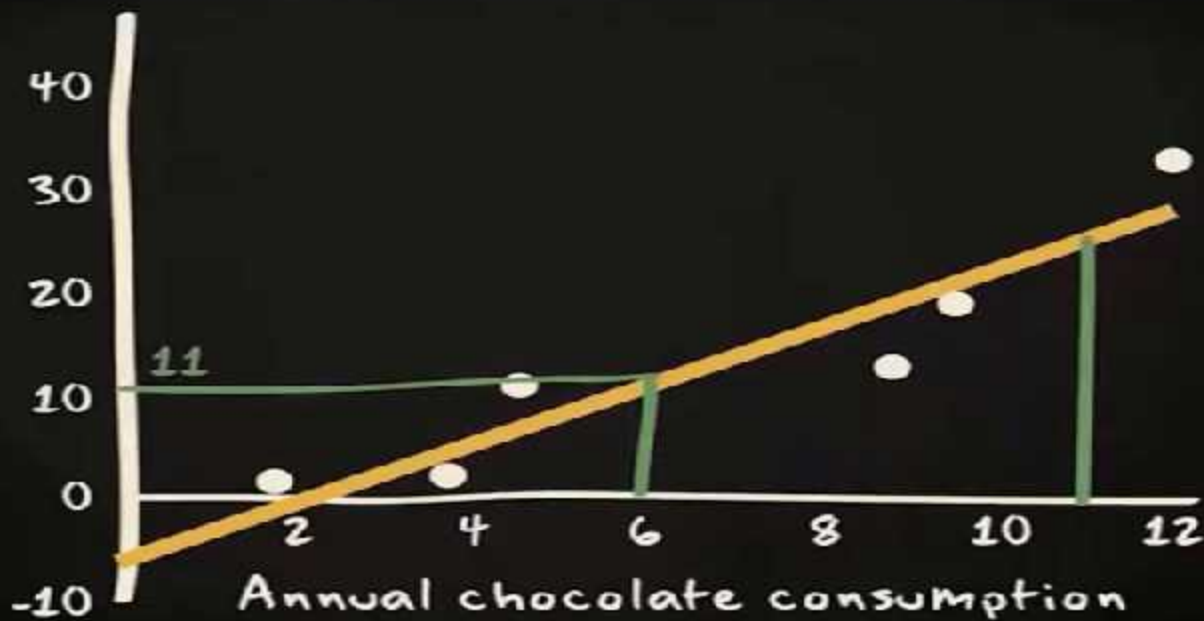
Number of Nobel prize winners
per 10 million people





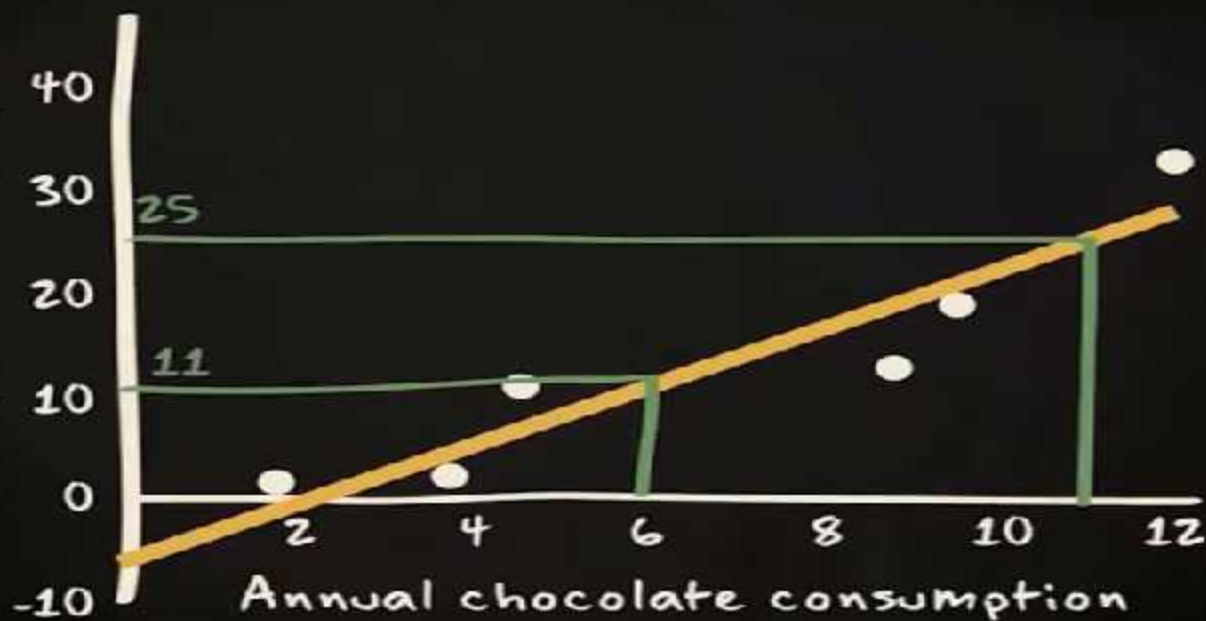
Number of Nobel prize winners

per 10 million people



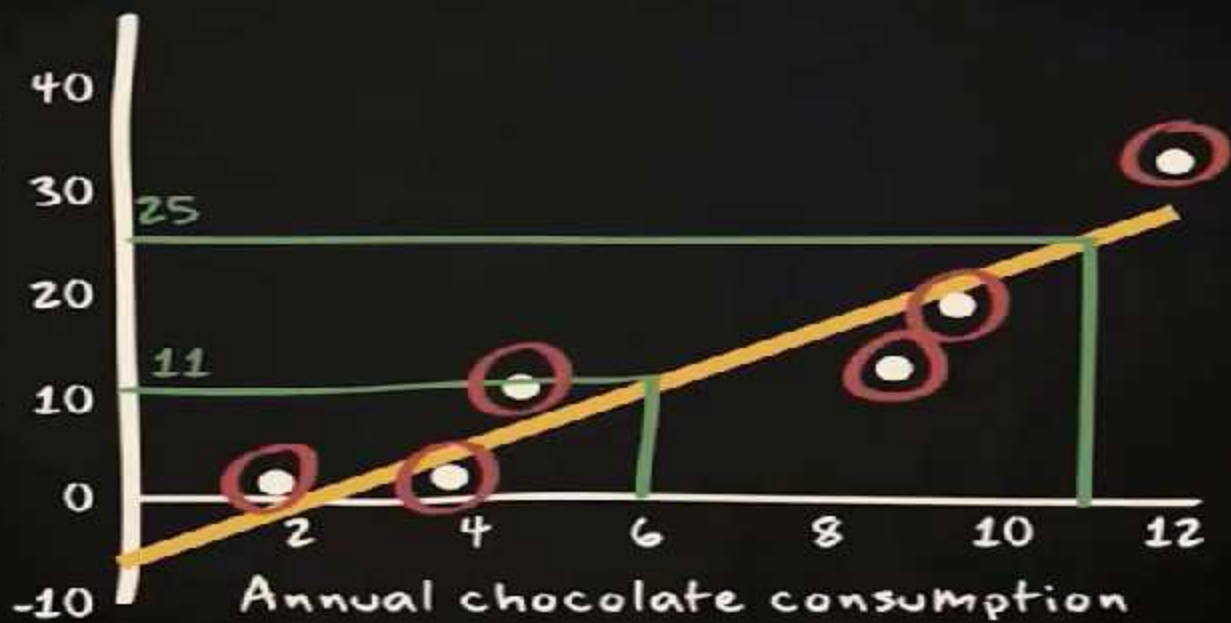


Number of Nobel prize winners
per 10 million people

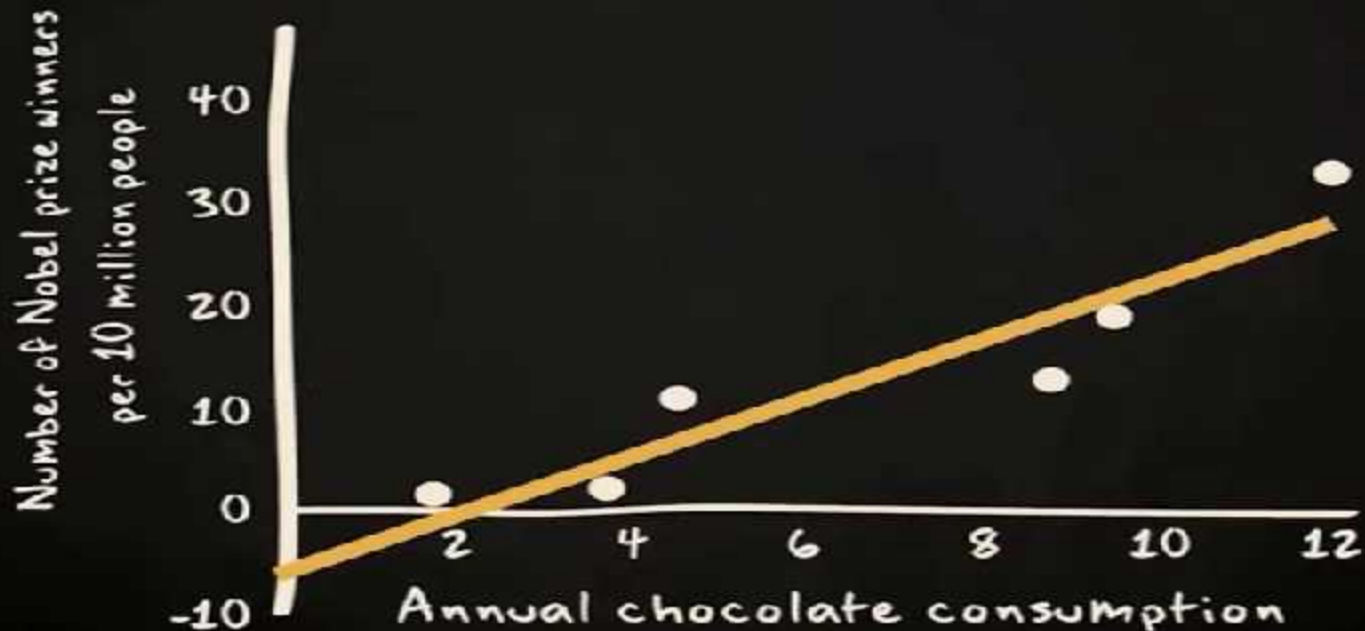




Number of Nobel prize winners
per 10 million people

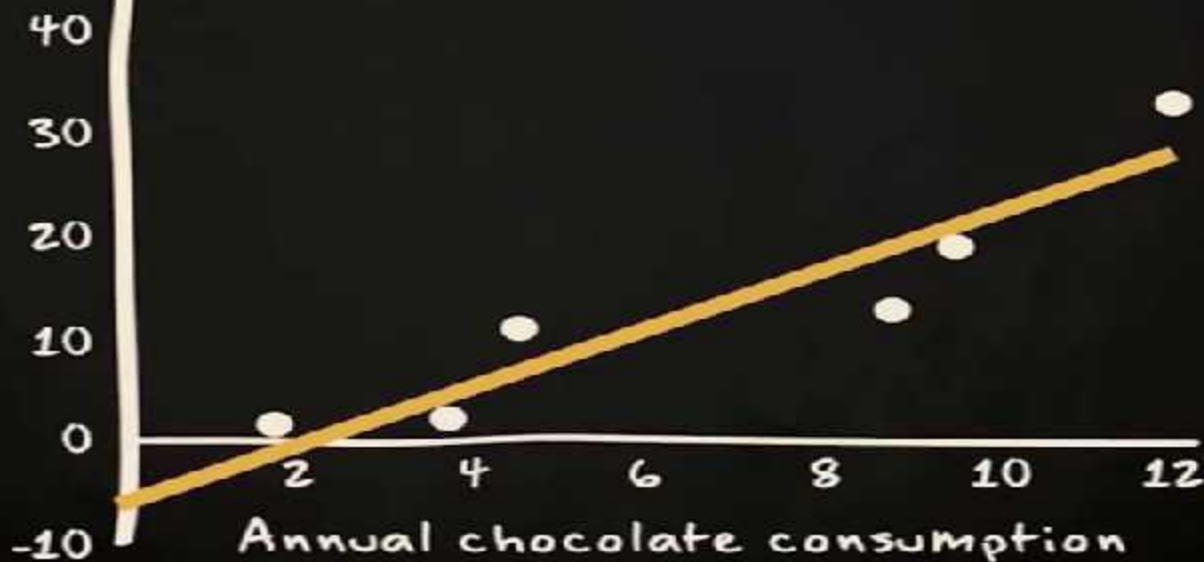


$$\hat{y} = a + bx$$





Number of Nobel prize winners
per 10 million people



$$\hat{y} = a + bx$$

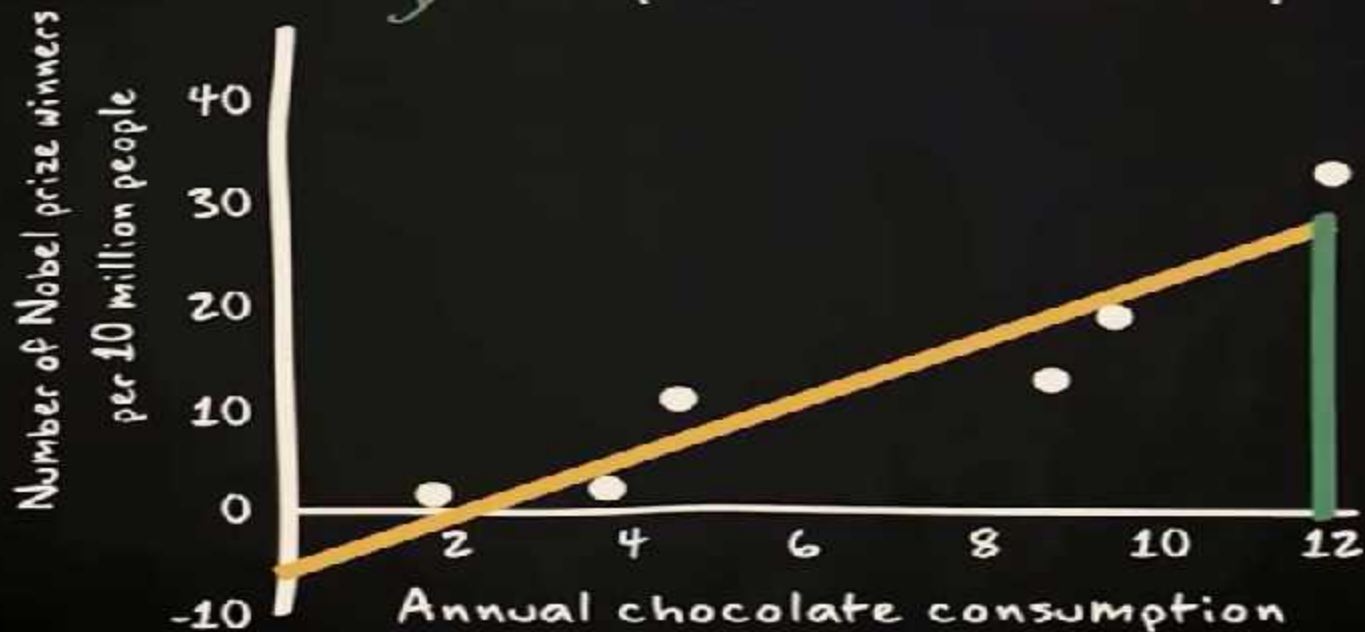
$\hat{y}$ = predicted value of y





$$\hat{y} = a + bx$$

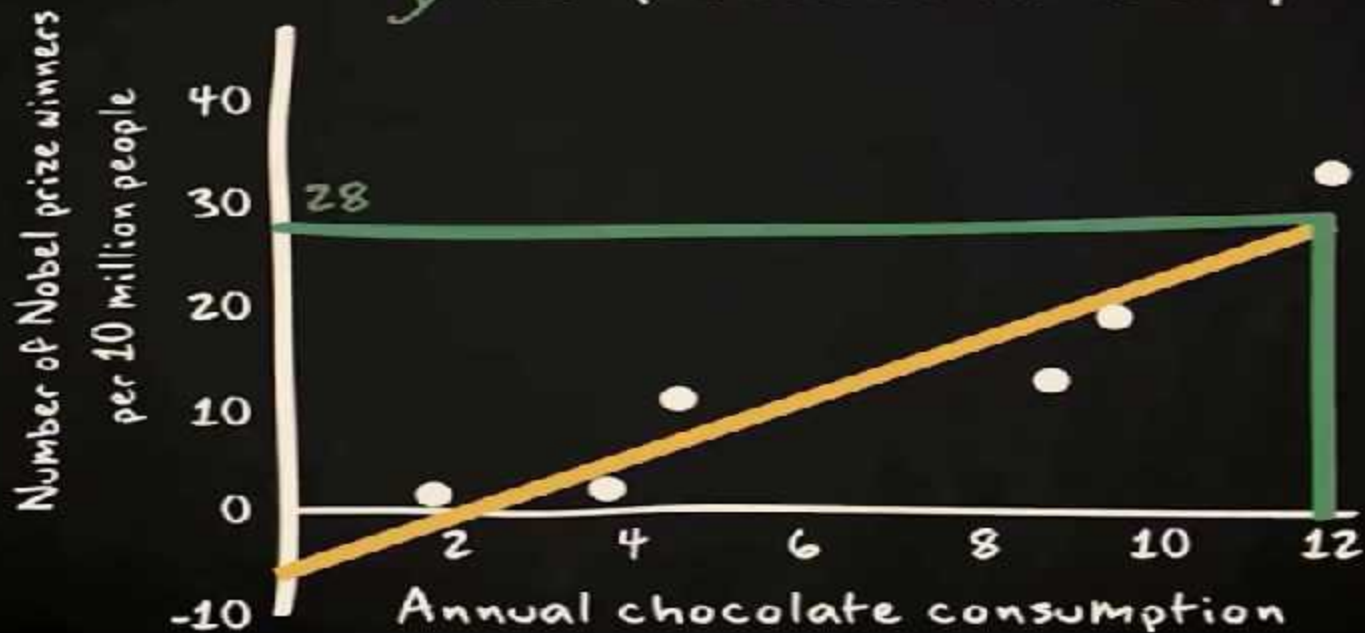
$\hat{y}$ = predicted value of y





$$\hat{y} = a + bx$$

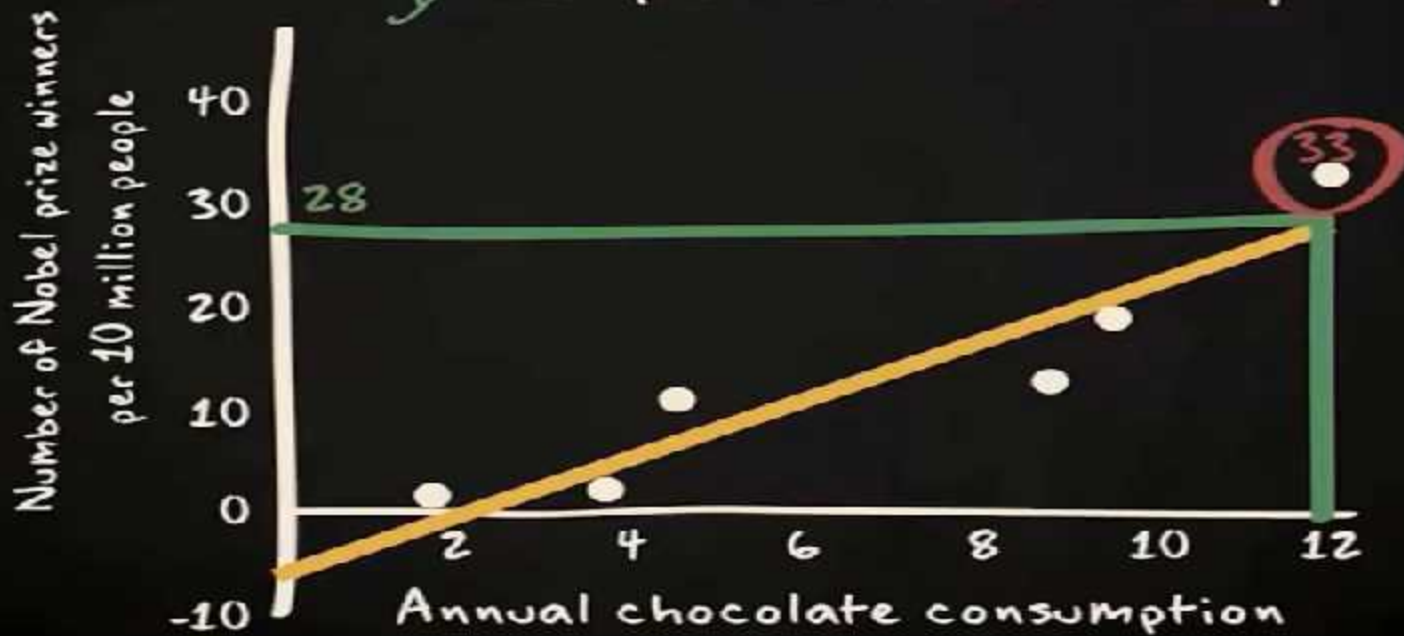
$\hat{y}$ = predicted value of y





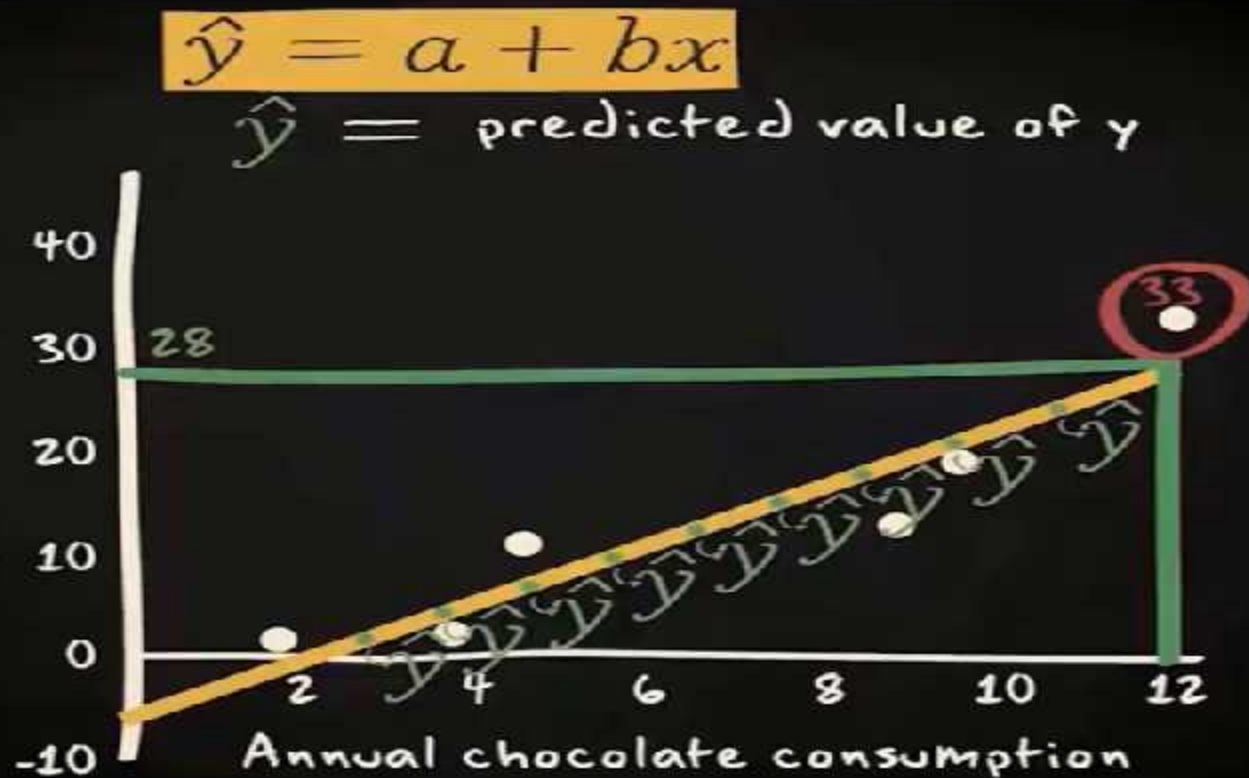
$$\hat{y} = a + bx$$

$\hat{y}$ = predicted value of y



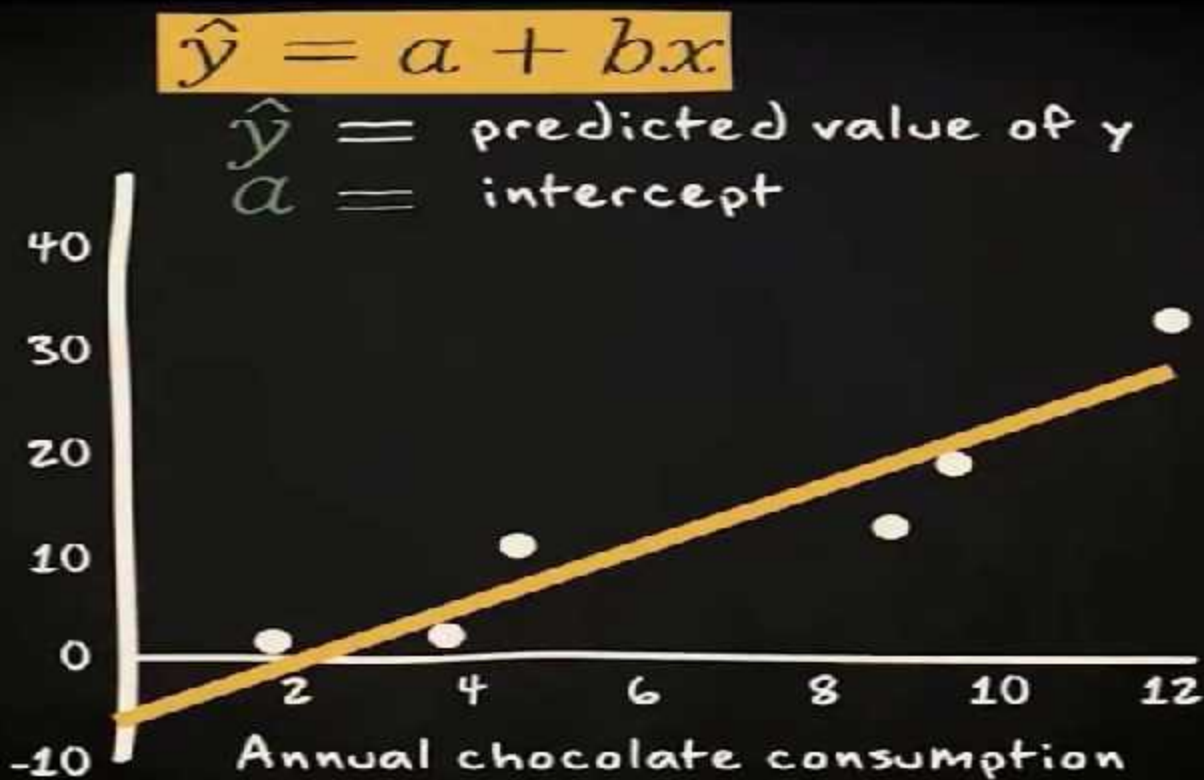


Number of Nobel prize winners
per 10 million people





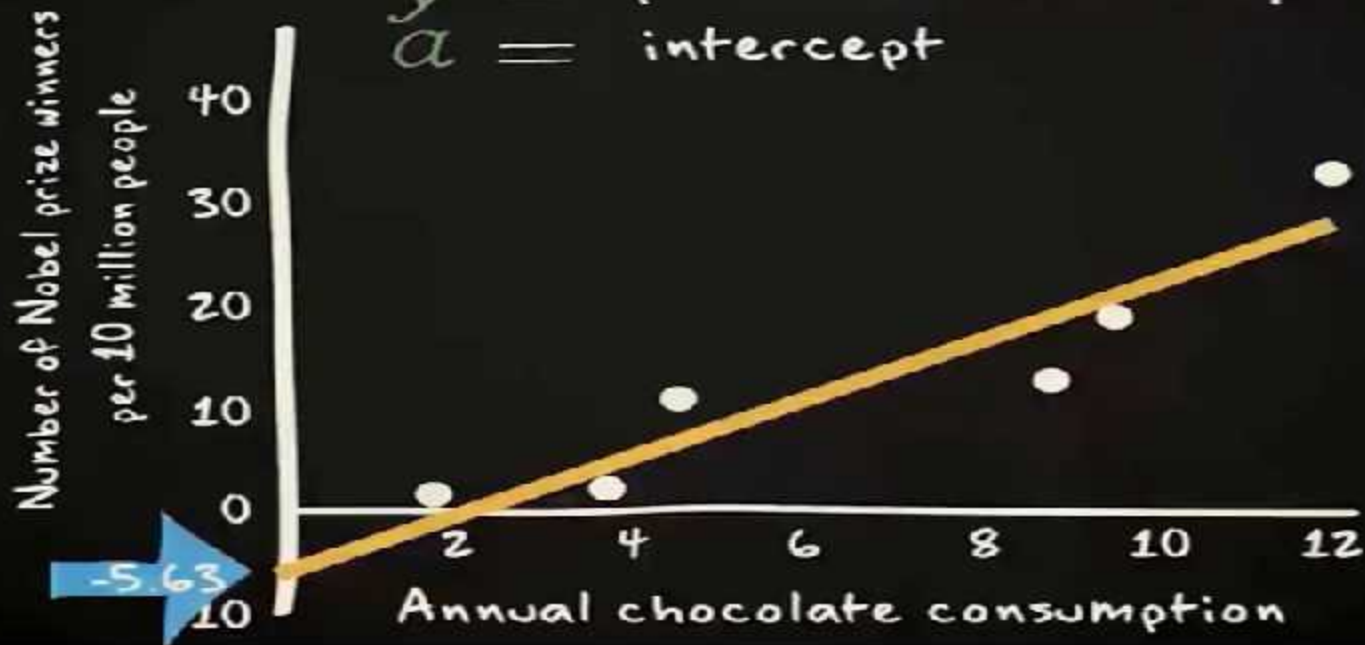
Number of Nobel prize winners
per 10 million people



$$\hat{y} = a + bx$$

$\hat{y}$ = predicted value of y

a = intercept





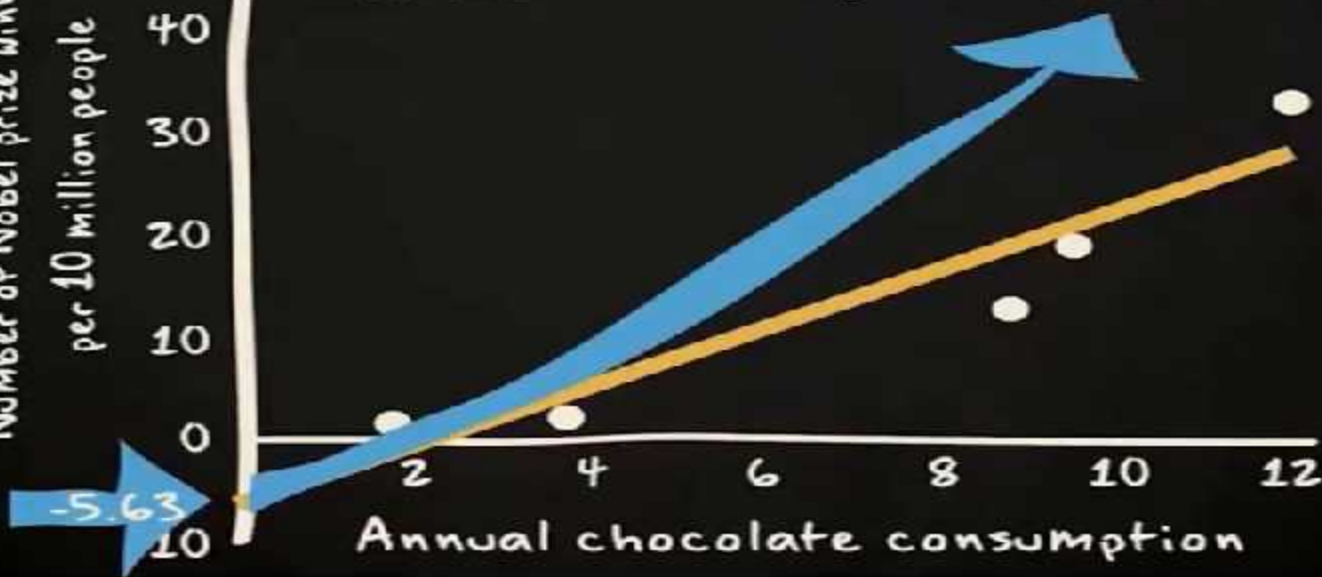
Number of Nobel prize winners
per 10 million people

40
30
20
10
0
-10

$$\hat{y} = a + bx$$

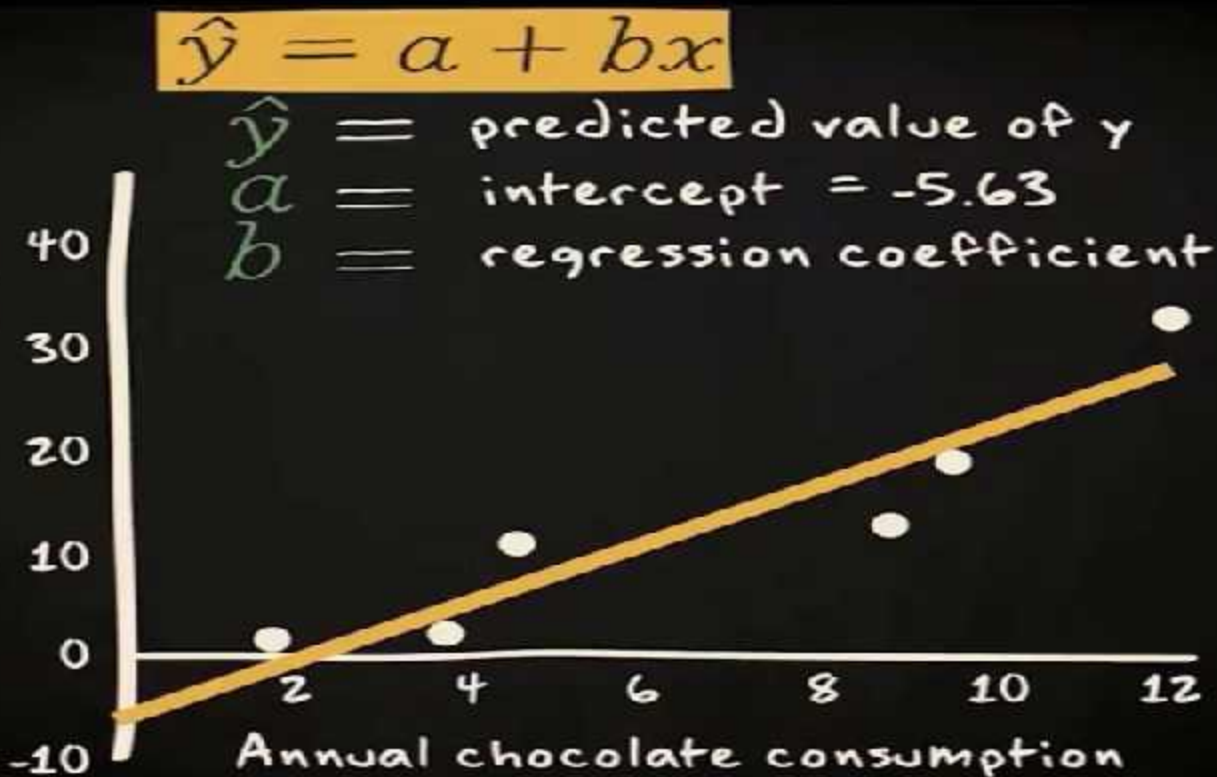
$\hat{y}$ = predicted value of y
 a = intercept = -5.63

Annual chocolate consumption



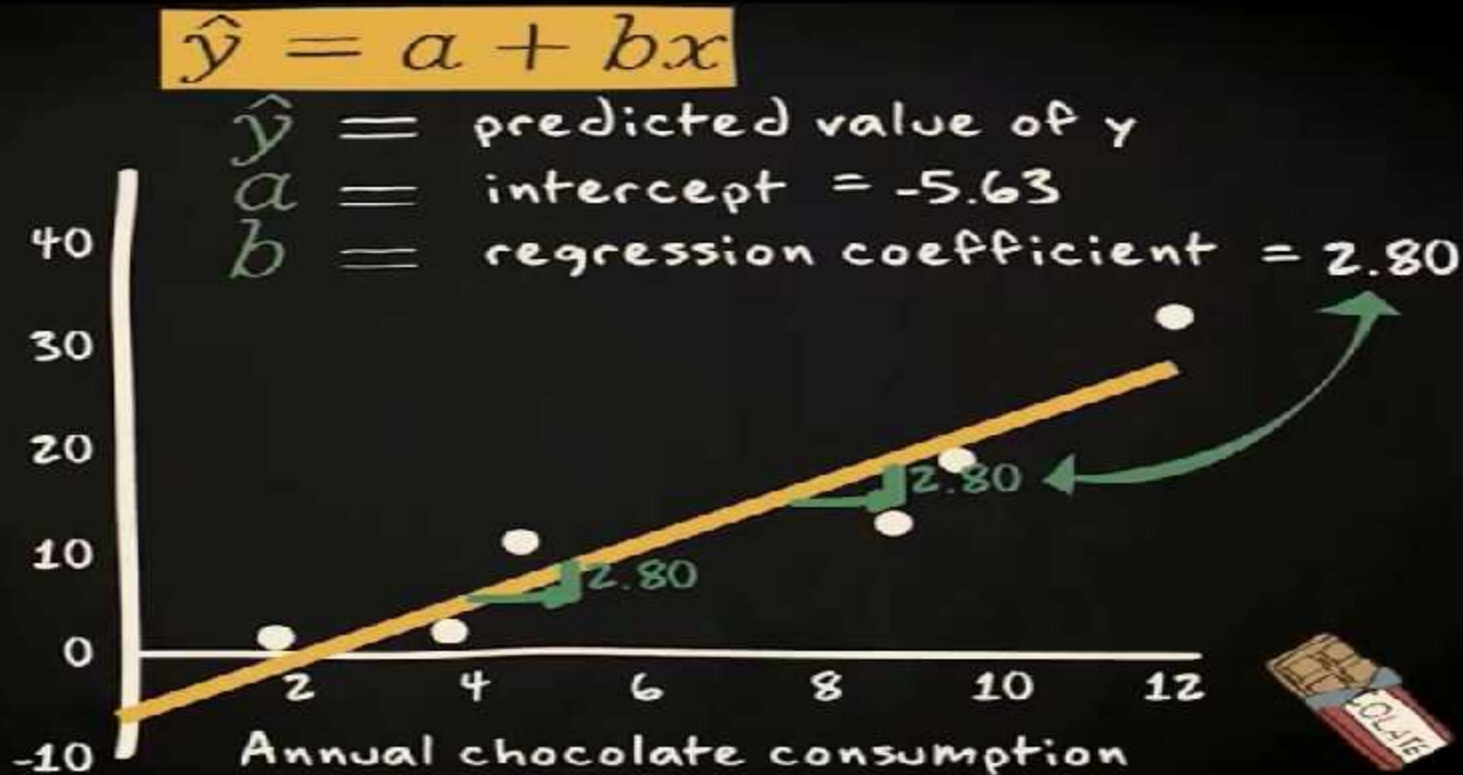


Number of Nobel prize winners
per 10 million people



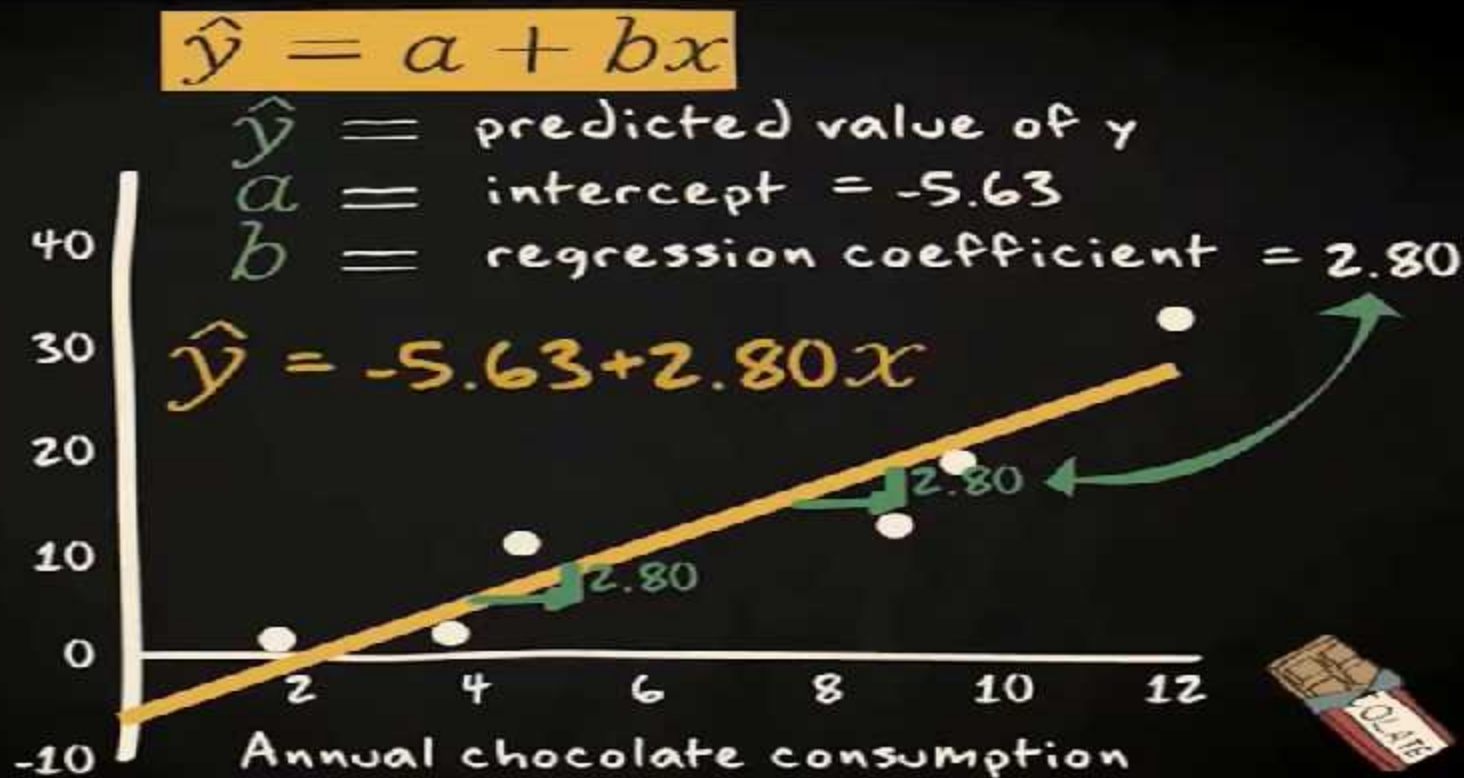


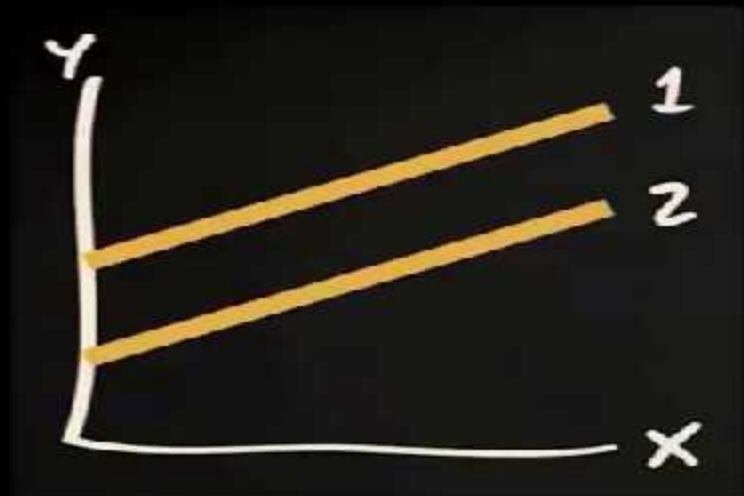
Number of Nobel prize winners
per 10 million people





Number of Nobel prize winners
per 10 million people







$$b_1 = b_2$$

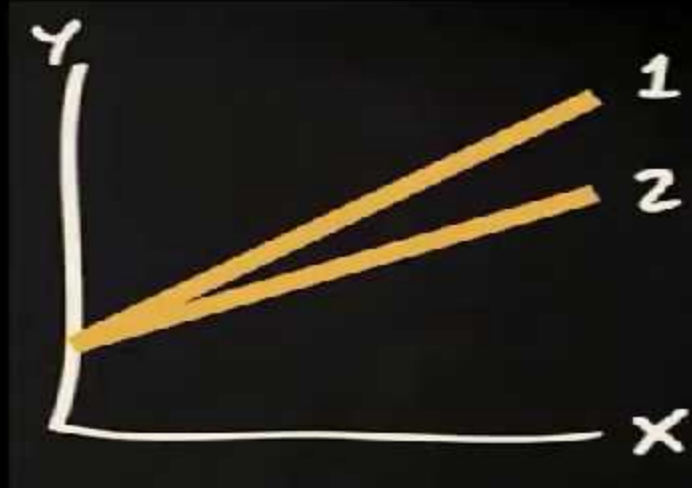


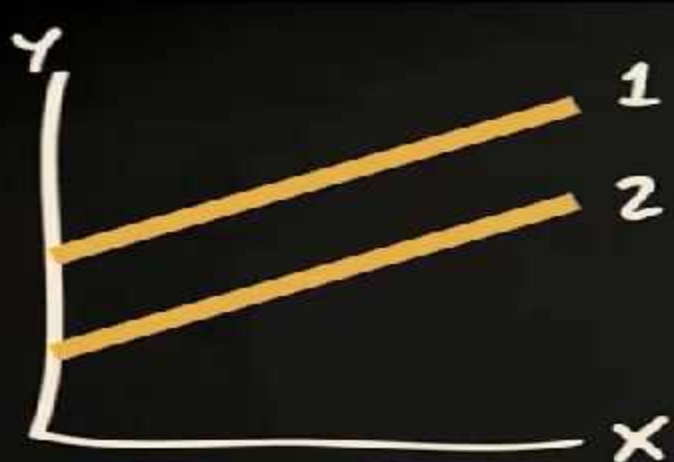
$$b_1 = b_2$$

$$a_1 \neq a_2$$

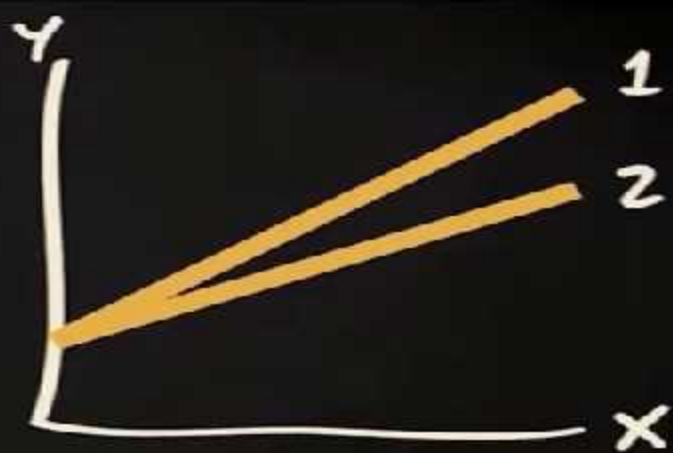


$$b_1 = b_2$$
$$a_1 \neq a_2$$





$$b_1 = b_2$$
$$a_1 \neq a_2$$



$$b_1 \neq b_2$$
$$a_1 = a_2$$

regression line

predict

y

regression line

predict

$\hat{Y}$

regression formula

predict

$\hat{Y}$

regression formula

predict

Y

$$\hat{y} = -5.63 + 2.80x$$

regression formula

predict

Y

$$\hat{y} = -5.63 + 2.80x$$

$$x = 3.5$$

$$\hat{y} = -5.63 + (2.80 * 3.5)$$

$$\hat{y} = 4.17$$

regression formula

predict

Y

$$\hat{y} = -5.63 + 2.80x$$

$$x = 3.5$$

$$\hat{y} = -5.63 + (2.80 * 3.5)$$

$$\hat{y} = 4.17$$

$$x = 10.21$$

$$\hat{y} = -5.63 + (2.80 * 10.21)$$

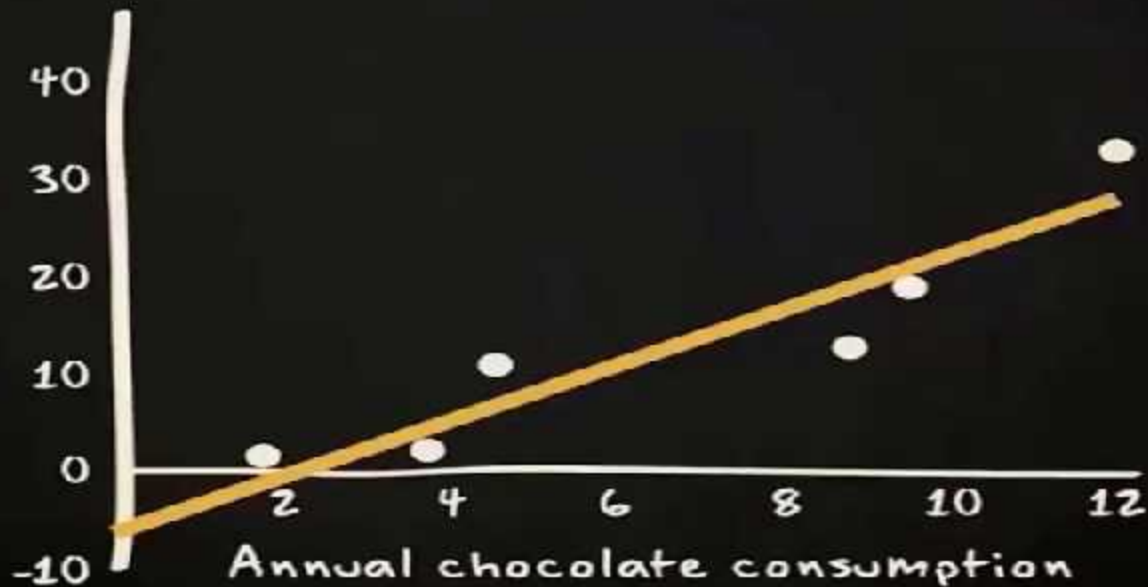
$$\hat{y} = 22.96$$

$$x = 3.5 \quad \hat{y} = 4.17$$

$$x = 10.21 \quad \hat{y} = 22.96$$



Number of Nobel prize winners
per 10 million people

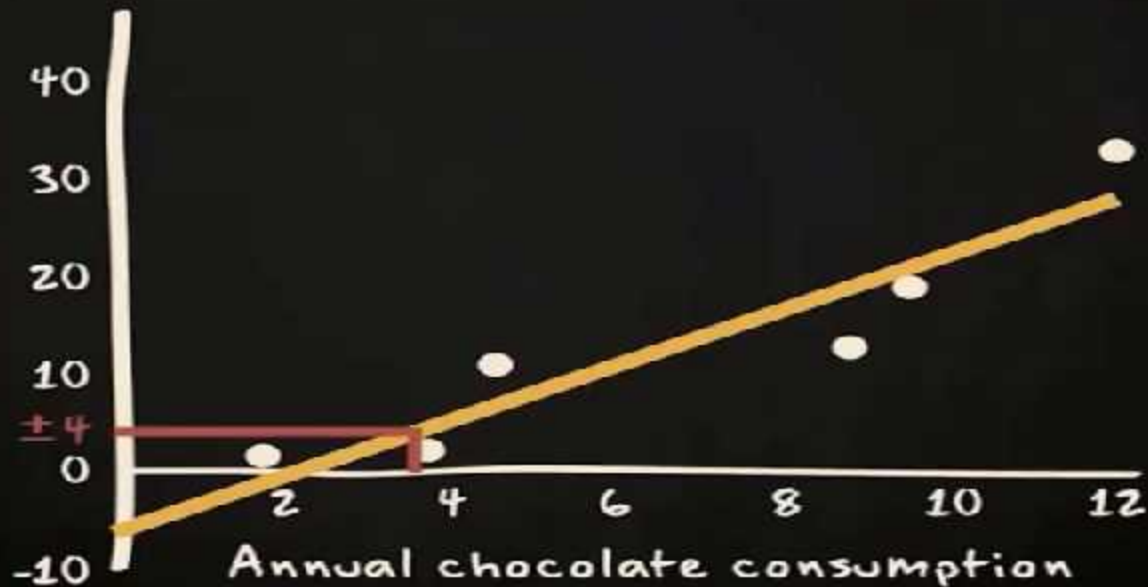


$$x = 3.5 \quad \hat{y} = 4.17$$

$$x = 10.21 \quad \hat{y} = 22.96$$



Number of Nobel prize winners
per 10 million people

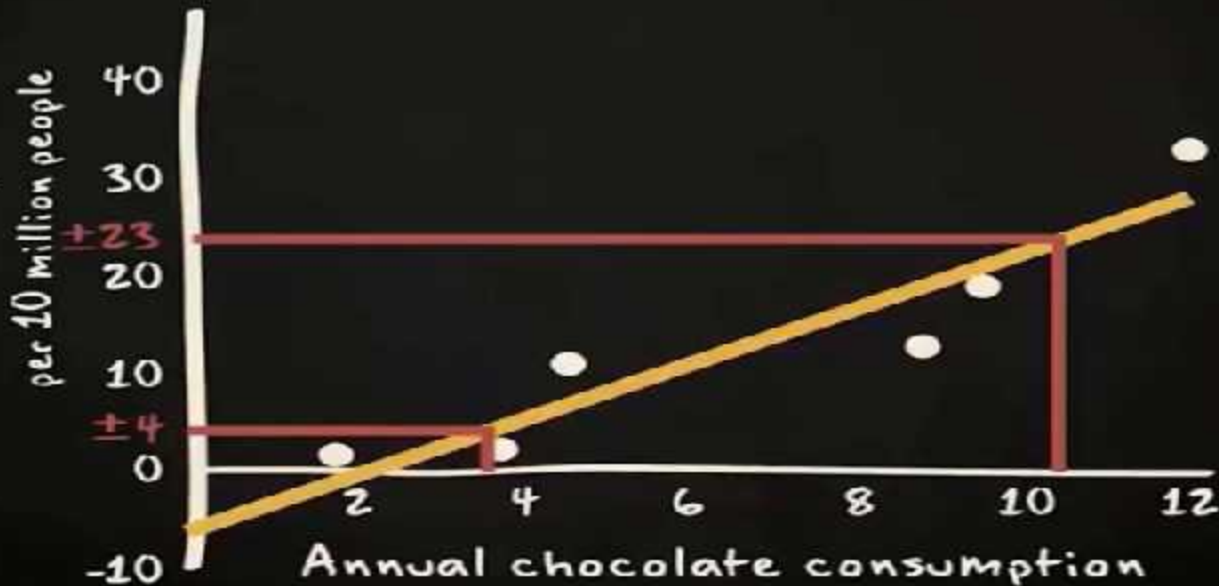


$$x = 3.5 \quad \hat{y} = 4.17$$

$$x = 10.21 \quad \hat{y} = 22.96$$



Number of Nobel prize winners



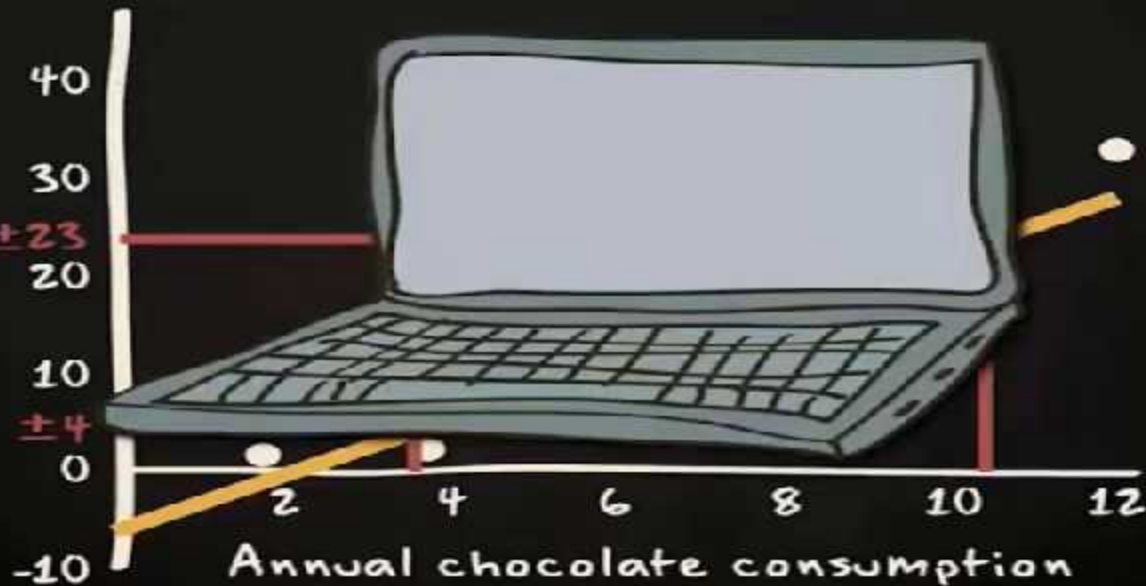
$$x = 3.5 \quad \hat{y} = 4.17$$

$$x = 10.21 \quad \hat{y} = 22.96$$



Number of Nobel prize winners

per 10 million people



COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression coefficient



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regression coefficient
unstandardized Pearson's r

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression coefficient
unstandardized Pearson's r
 $r = 0, b = 0$

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression coefficient
unstandardized Pearson's r
 $r = 0, b = 0$
 $r = +$

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression coefficient
unstandardized Pearson's r

$$r = 0, b = 0$$

$$r = +, b = +$$

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

regression coefficient
unstandardized Pearson's r

$$r = 0, b = 0$$

$$r = +, b = +$$

$$r = -, b = -$$

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

$$\bar{x} = 6.71$$

$$\bar{y} = 13.17$$

$$s_x = 3.95$$

$$s_y = 11.87$$

$$r = 0.93$$

COMPUTE REGRESSION LINE

$$b = r \frac{s_y}{s_x}$$

$$a = \bar{y} - b(\bar{x})$$

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$$s_x = 3.95$$

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COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

$$b = 0.93 \left(\frac{11.87}{3.95} \right)$$

$$\bar{x} = 6.71$$

$$\bar{y} = 13.17$$

$$s_x = 3.95$$

$$s_y = 11.87$$

$$r = 0.93$$

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

$$b = 0.93 \left(\frac{11.87}{3.95} \right)$$

$$b = 2.79$$

$$\bar{x} = 6.71$$

$$\bar{y} = 13.17$$

$$s_x = 3.95$$

$$s_y = 11.87$$

$$r = 0.93$$

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

$$b = 0.93 \left(\frac{11.87}{3.95} \right)$$

$$b = 2.79$$

$$\bar{x} = 6.71$$

$$\bar{y} = 13.17$$

$$s_x = 3.95$$

$$s_y = 11.87$$

$$r = 0.93$$

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b \bar{x}$$

$$b = 0.93 \cdot \left(\frac{11.87}{3.95} \right)$$

$$b = 2.79$$

$$\bar{x} = 6.71$$

$$\bar{y} = 13.17$$

$$s_x = 3.95$$

$$s_y = 11.87$$

$$r = 0.93$$

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

$$b = 0.93 \cdot \left(\frac{11.87}{3.95} \right) \quad a = 13.17 - (2.79 \cdot 6.71)$$

$$b = 2.79$$

$$\bar{x} = 6.71$$

$$\bar{y} = 13.17$$

$$s_x = 3.95$$

$$s_y = 11.87$$

$$r = 0.93$$

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

$$b = 0.93 \left(\frac{11.87}{3.95} \right)$$

$$a = 13.17 - (2.79 \cdot 6.71)$$

$$b = 2.79$$

$$a = -5.55$$

$$\bar{x} = 6.71$$

$$\bar{y} = 13.17$$

$$s_x = 3.95$$

$$s_y = 11.87$$

$$r = 0.93$$

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

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$$s_y = 11.87$$

$$r = 0.93$$

$$\hat{y} = a + bx$$

$$\hat{y} = -5.55 + 2.79x$$

COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

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$$b = 2.79$$

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$$r = 0.93$$

$$\hat{y} = a + bx$$

$$\hat{y} = -5.55 + 2.79x$$

$$\hat{y} = -5.63 + 2.80x$$



COMPUTE REGRESSION LINE

$$b = r \left(\frac{s_y}{s_x} \right)$$

$$a = \bar{y} - b(\bar{x})$$

$$b = 0.93 \left(\frac{11.87}{3.95} \right) \quad a = 13.17 - (2.79 \cdot 6.71)$$

$$b = 2.79$$

$$a = -5.55$$

$$\bar{x} = 6.71$$

$$\bar{y} = 13.17$$

$$s_x = 3.95$$

$$s_y = 11.87$$

$$r = 0.93$$

$$\hat{y} = a + bx$$

$$\hat{y} = -5.55 + 2.79x$$

$$\hat{y} = -5.63 + 2.80x$$

rounding error

